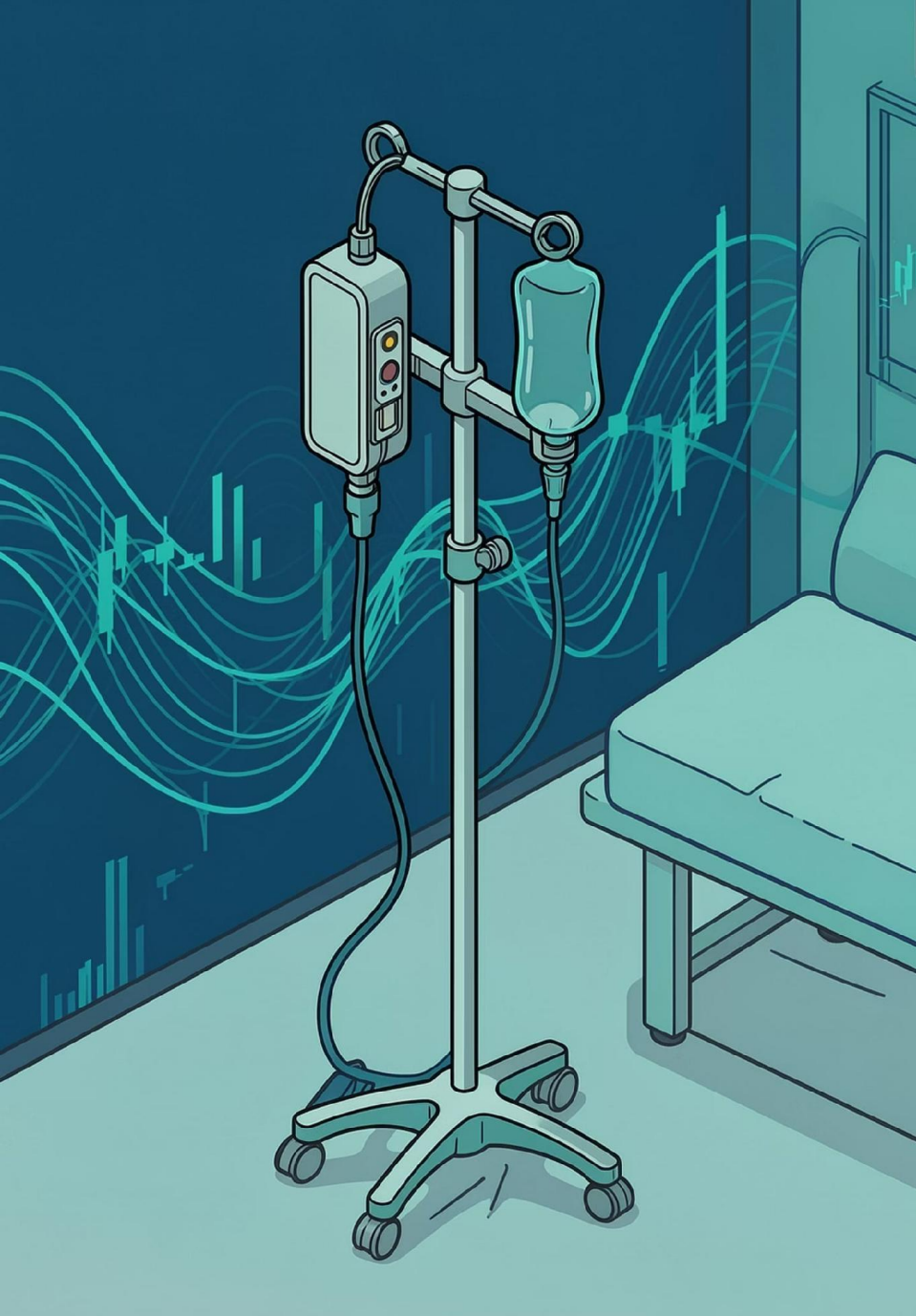




AI-Driven IV Fluid Monitoring and Smart Alert System

Proactive Smart Alerts for Resource-Constrained Healthcare

Innovation in adoption of Industry 4.0 & 5.0 for MSME healthcare — Medical Devices & Healthcare IoT

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Problem Statement & Objectives



The Problem

- Manual IV monitoring lets empty bags go unnoticed → increased risk of air embolism.
 - Current systems are reactive: loud ward alarms that contribute to alarm fatigue and disrupted care.
 - Proprietary pumps and cloud-dependent systems are costly and unsuitable for under-resourced MSME hospitals.
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Objectives

- Predictive Safety — detect anomalies before a critical event.
- Targeted Alerts — replace loud buzzers with silent, mobile triage notifications.
- Cost-Effective — retrofit approach to bring smart monitoring to MSME hospitals.

Proposed Solution vs Existing Systems

Feature	Existing Systems	Our Innovation
Detection	Reactive — triggers only when IV is empty	Predictive — Edge AI anticipates empty/flow issues
Alert Type	Loud ward buzzers (disruptive)	Silent mobile push notifications to triage
Cost	High — proprietary pumps and licensing	Low — retrofit IoT node using off-the-shelf parts
Latency	High — cloud-dependent delays	Near-zero — on-device Edge computing

Five-Step Workflow (Process Flow)



1. Sense

Load cell measures fluid weight continuously on the IV pole.



2. Amplify

HX711 converts analog load-cell output to precise digital readings.



3. Analyze

ESP32 ingests time-series data and runs TinyML models locally.



4. Predict

AI classifies flow: Normal, Blocked, or Rapidly Emptying — early anomaly detection.

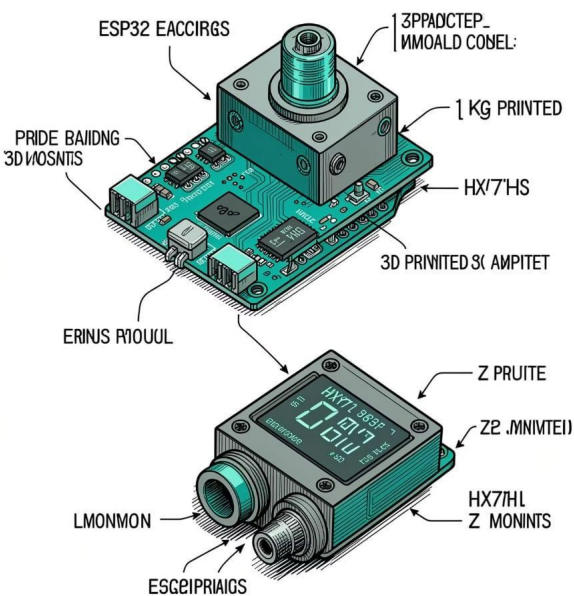


5. Alert

Blynk cloud relays targeted silent pushes to nursing mobile apps — avoids ward-wide buzzers.

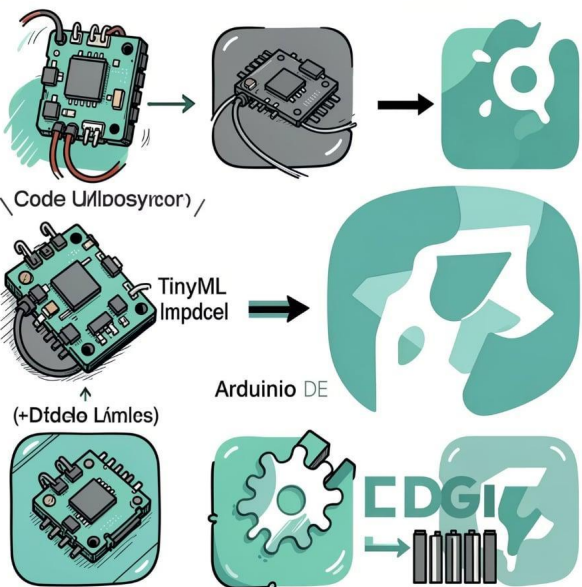
System Architecture – Three Pillars

Hardware



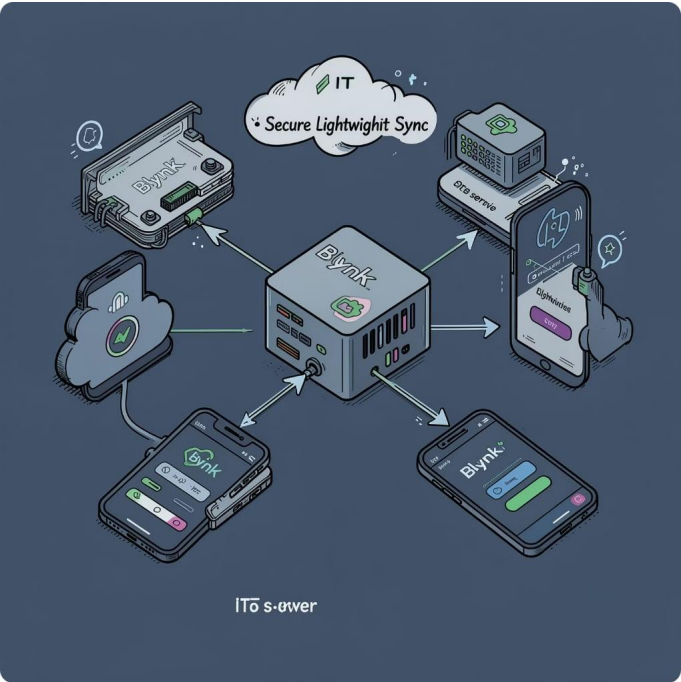
ESP32 microcontroller • 1kg Load Cell • HX711 • 3D printed mounts for universal retrofit

Software



C++ (Arduino) for firmware • TinyML models via Edge Impulse • Local inference pipeline

Cloud & Mobile



Blynk IoT server for device management & mobile push — mobile UI for nurse triage (iOS/Android)

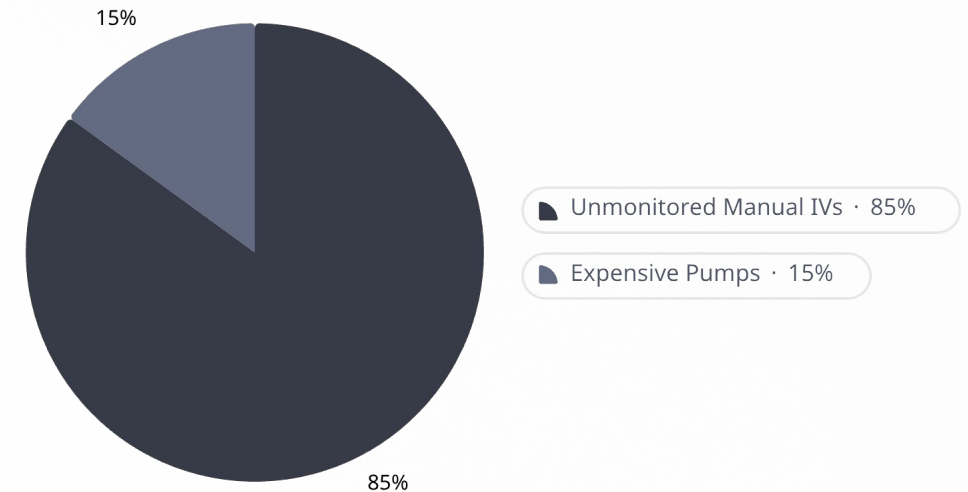
Results & Market Analysis

Expected Clinical Impact

- Near-elimination of air-embolism events by early detection and intervention.
- Up to 40% reduction in ward alarm noise through targeted mobile alerts.
- Enables affordable, scalable monitoring for Tier-2/Tier-3 hospitals and rural clinics.

Market Opportunity

Large addressable market in under-resourced MSME hospitals where expensive smart pumps are unaffordable. Retrofit strategy lowers barrier to adoption and accelerates impact.



Current Market: 85% manual IVs remain unmonitored — a clear entry point for low-cost edge solutions.

Chart explains market composition and highlights the workflow: sensing → edge inference → mobile triage → clinical intervention.

Innovation, SDGs & Future Scope

Novelty

Patentable approach: first localized Edge AI on low-cost load-cell microcontrollers for predictive IV monitoring.

SDGs

Targets SDG 3 (Good Health & Well-being) and SDG 9 (Industry, Innovation & Infrastructure).

Future Scope

EMR integration, AI scaling to detect transfusion viscosities, multi-parameter monitoring (flow + pressure).



Planned next steps include clinical pilots, regulatory pathway mapping, and partnerships with MSME hospital networks.

Conclusion



Conclusion

We convert common IV equipment into predictive Industry 4.0 infrastructure: protecting patients, reducing nurse burden, and keeping costs within MSME budgets.



Questions?

THANK YOU!